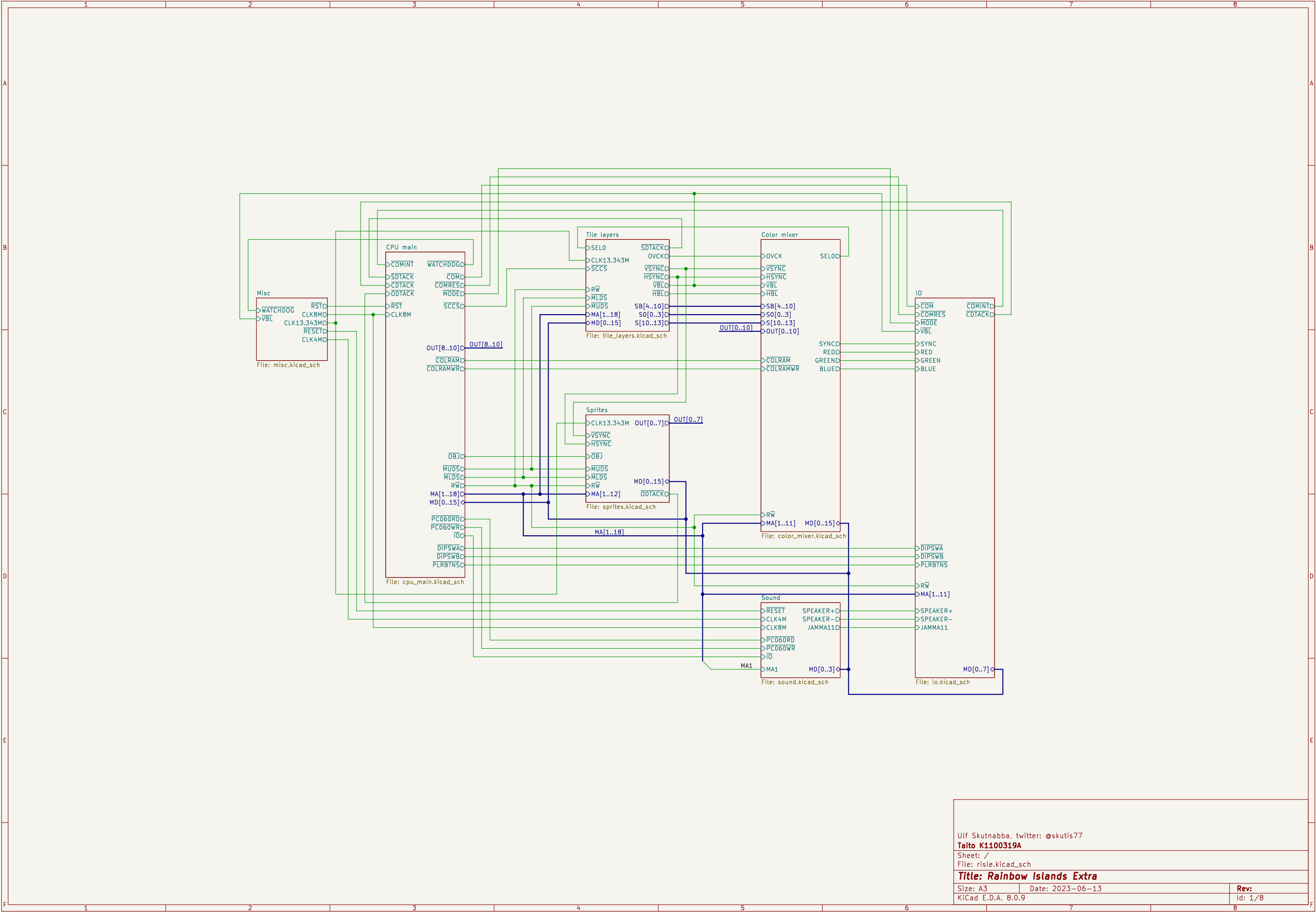




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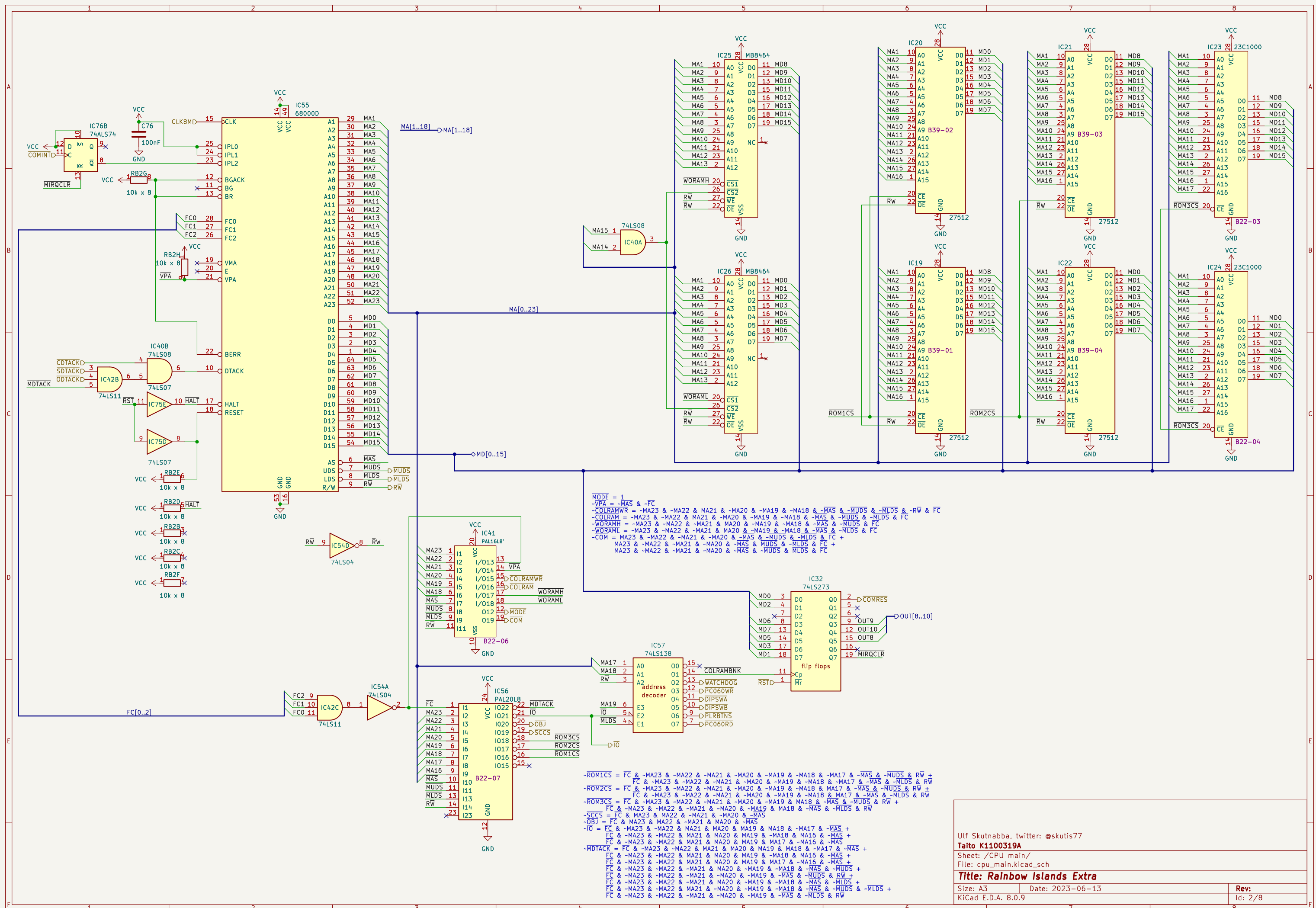
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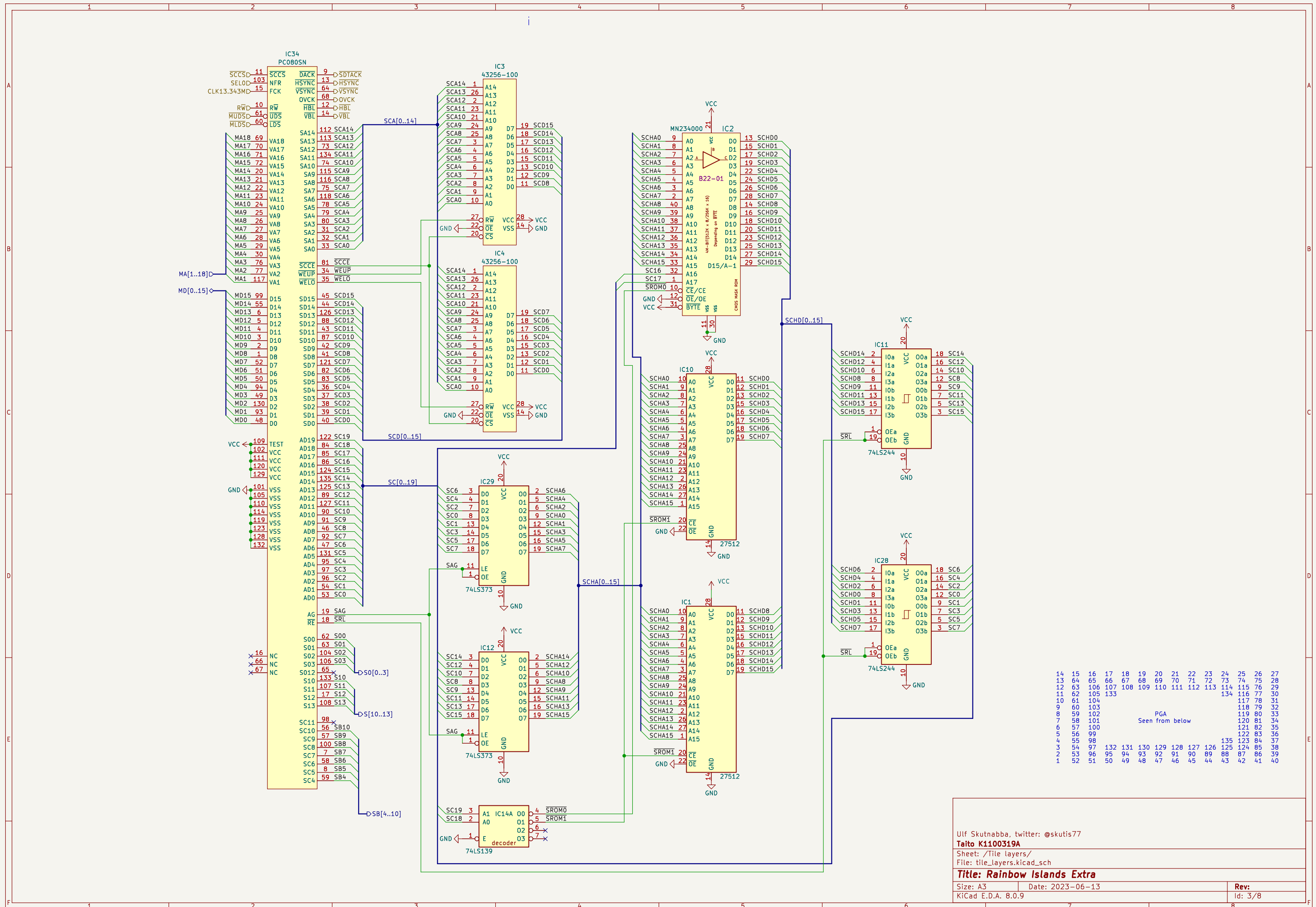
Sheet: /
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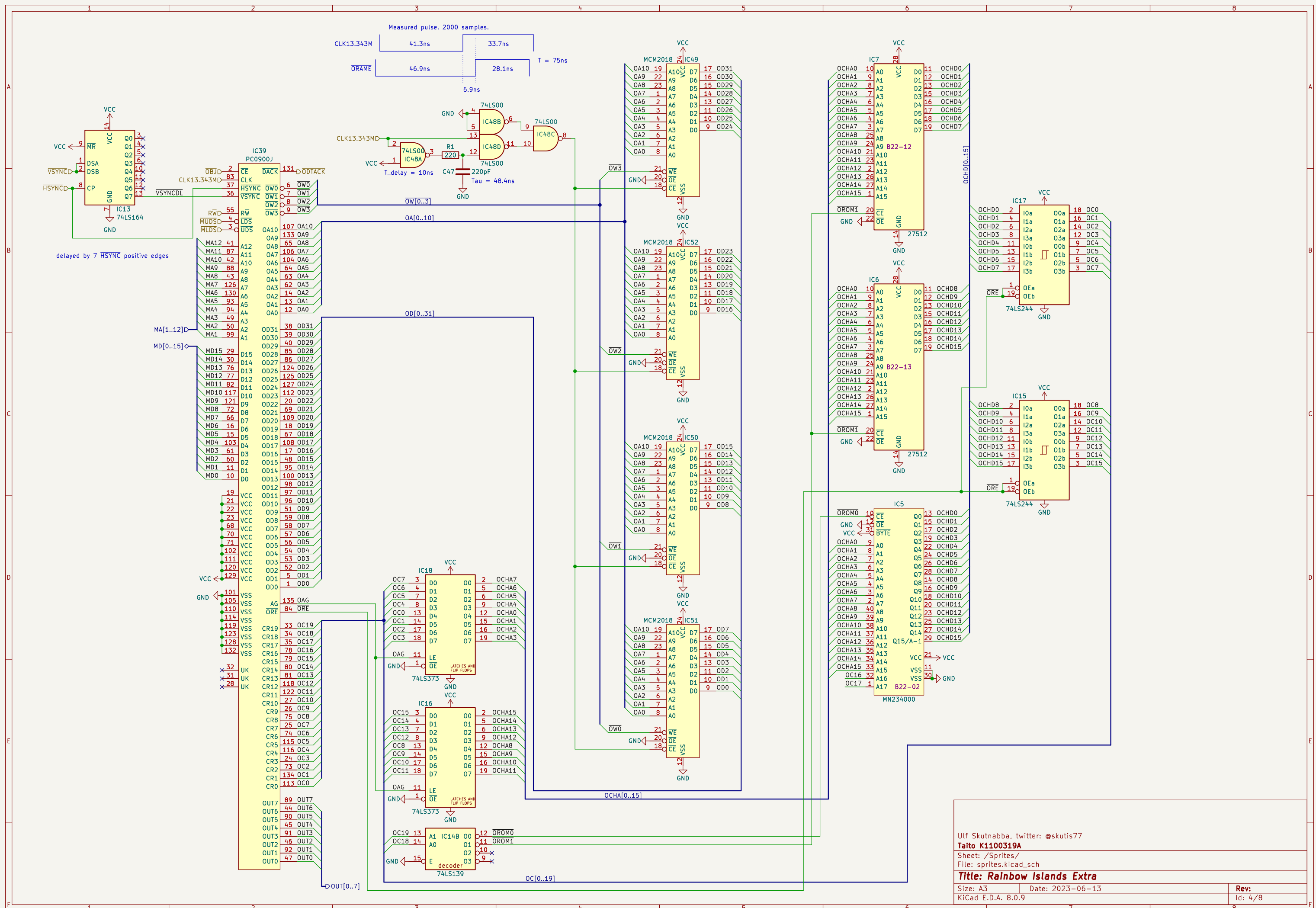
Title: Rainbow Islands Extra

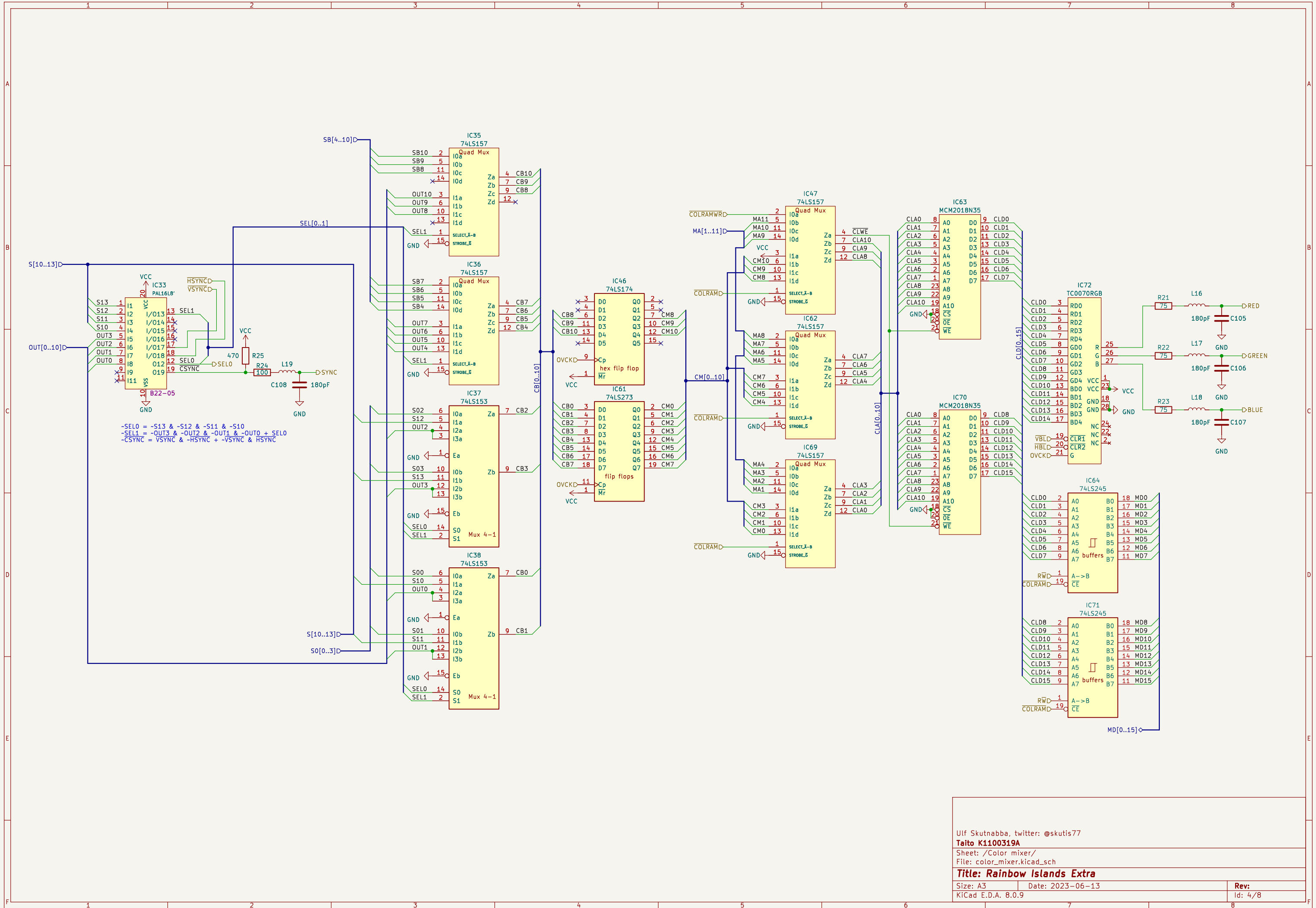
Size: A3 Date: 2023-06-13
KiCad E.D.A. 8.0.9

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Id: 1/8









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Sheet: /Color mixer/

File: color_mixer.kicad_sch

Title: Rainbow Islands Extra

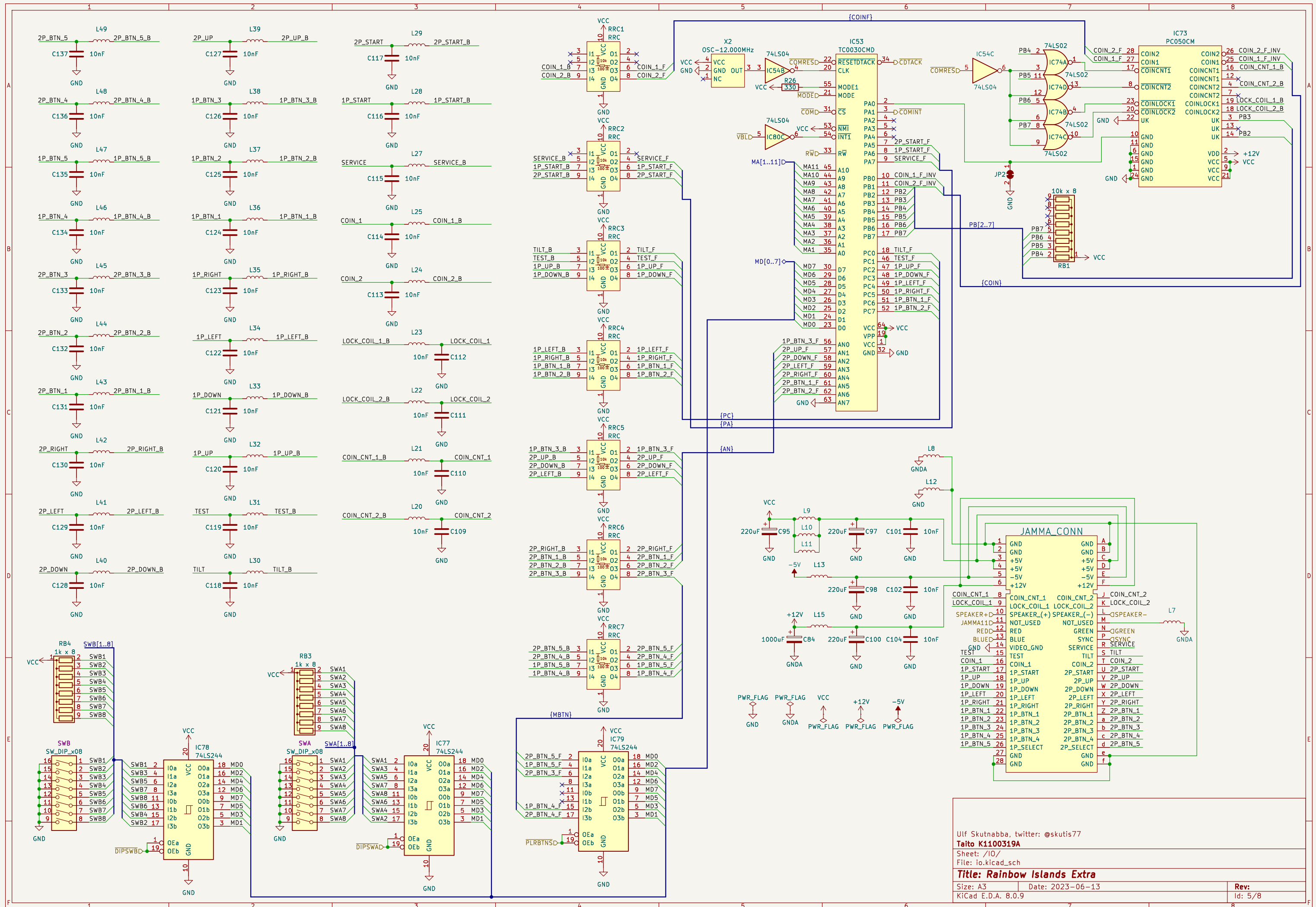
Size: A3

Date: 2023-06-13

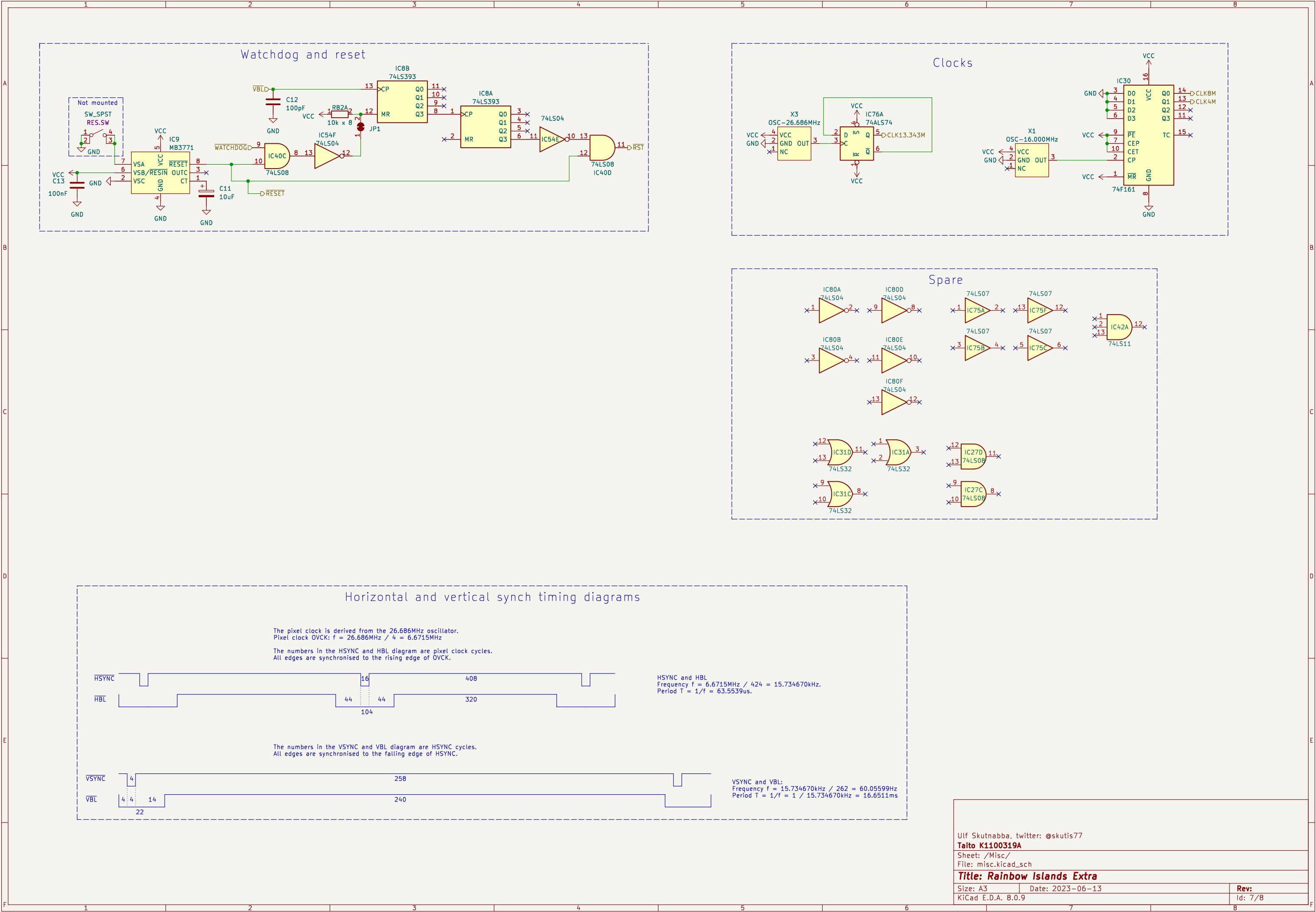
Rev:

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Id: 4/8







Horizontal and vertical synch timing diagrams

The pixel clock is derived from the 26.686MHz oscillator.
Pixel clock OVCK: $f = 26.686\text{MHz} / 4 = 6.6715\text{MHz}$

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of OVCK.

HSYNC

HBL

16

408

44

44

320

104

HSYNC and HBL
Frequency $f = 6.6715\text{MHz} / 424 = 15.734670\text{kHz}$.
Period $T = 1/f = 63.5539\mu\text{s}$.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.

VSYNC

VBL

4

258

4

14

240

22

VSYNC and VBL:
Frequency $f = 15.734670\text{kHz} / 262 = 60.05599\text{Hz}$
Period $T = 1/f = 1 / 15.734670\text{kHz} = 16.6511\text{ms}$

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Sheet: /Misc/
File: misc.kicad_sch

Title: Rainbow Islands Extra

Size: A3
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Date: 2023-06-13

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Id: 7/8